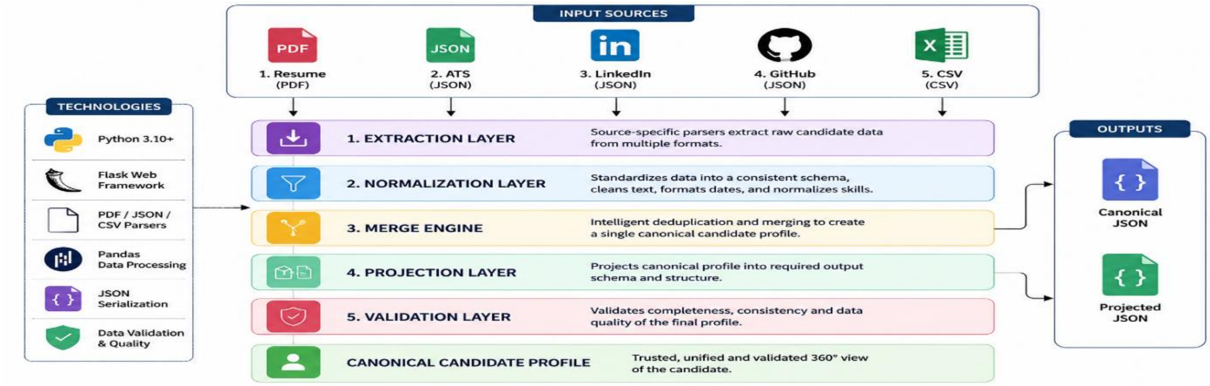
CandidateOne Technical Design

End-to-End Multi-Source Candidate Canonicalization Engine (Multi-Source Candidate Data Transformer)



1. Objective

CandidateOne is a deterministic ETL pipeline that transforms fragmented **candidate information from multiple structured and unstructured sources into a single validated canonical candidate profile.** The system emphasizes consistency, explainability, provenance tracking, confidence scoring, and configurable output generation while ensuring identical inputs always produce identical outputs.

2. Pipeline Design

Source Detection (**Unstructured** – resume.pdf, **Structured** – JSON files) → Extraction → Normalization → Merge & Conflict Resolution → Projection → Validation → **Canonical / Projected Output**.

3. Canonical Candidate Schema

The pipeline converts every input source into a unified canonical candidate representation before any downstream processing begins. Regardless of whether the data originates from a resume, ATS export, LinkedIn profile, GitHub profile, or recruiter CSV, each source is normalized into the same internal schema. This canonical schema includes essential candidate attributes such as the Candidate ID, Full Name, Email Addresses, Phone Numbers, Location, Professional Links, Headline, Years of Experience, Skills, Experience, Education, Provenance, and Overall Confidence Score. By standardizing all incoming data into a common structure, the pipeline isolates downstream components from source-specific formats, simplifies merge and validation logic, and ensures consistent, reliable, and extensible processing throughout the system.

4. Merge & Conflict Resolution Strategy

CandidateOne deterministically merges all extracted profiles into a single trusted candidate record.
Conflict resolution follows deterministic rules: Prefer non-empty values, Prefer higher-confidence sources, Preserve provenance for every field, Remove duplicate skills and records, Never invent missing information,
Default source priority: Resume → LinkedIn → ATS → GitHub → Recruiter CSV

5. Projection, Validation & Reliability including Edge Cases

After merging, the canonical profile is reshaped into the required output using a configurable Projection Layer, which selects required fields, renames attributes, applies output normalization, and optionally includes provenance and confidence information. The projected profile is then validated against the target JSON schema before being returned. An Overall Confidence Score (0.0–1.0) is computed using source reliability, profile completeness, and validation success. Default source weights are: Resume (0.30), LinkedIn (0.25), ATS (0.20), GitHub (0.15), and Recruiter CSV (0.10). The pipeline is deterministic and fault-tolerant, gracefully handling missing or conflicting data, duplicate records, invalid emails or phone numbers, unknown fields, empty recruiter rows, and partial profiles, ensuring a valid canonical candidate profile is produced whenever possible.

CandidateOne safely handles: Missing candidate information, Duplicate candidate records,
Invalid phone numbers, Invalid email addresses, Empty recruiter rows, Conflicting source values,
Unknown JSON attributes, Partial profiles
Instead of failing, the pipeline **returns a valid canonical profile whenever possible.**

6. Design Principles

✓ Deterministic Processing ✓ Modular ETL Architecture ✓ Explainable Merge Logic ✓ Schema Validation
✓ Configurable Output Projection ✓ Provenance Tracking ✓ Scalable & Maintainable Design

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